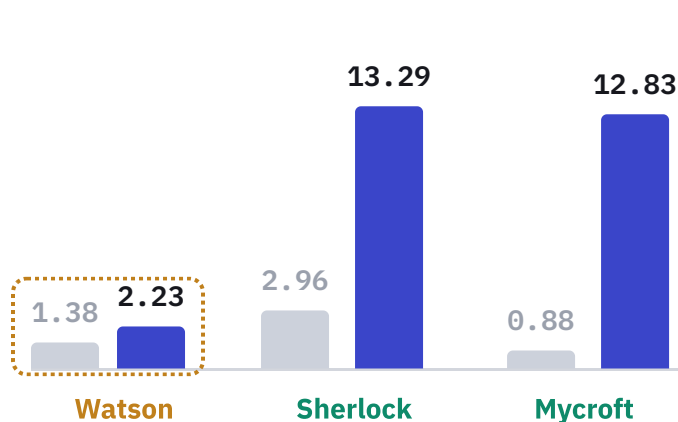


Low base competence: gains stay scaffold-local

Qwen3-4B · trained on Sherlock + Mycroft

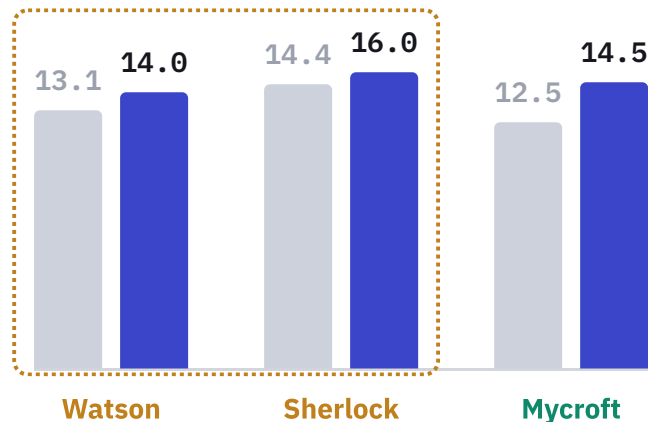


■ Base model ■ Post-trained model

RL lifts the **trained** scaffolds **4–15×**, but the **held-out** Watson scaffold, which demands internal game knowledge, **barely improves**.

High base competence: gains transfer everywhere

GPT-OSS-120B · trained on Mycroft only

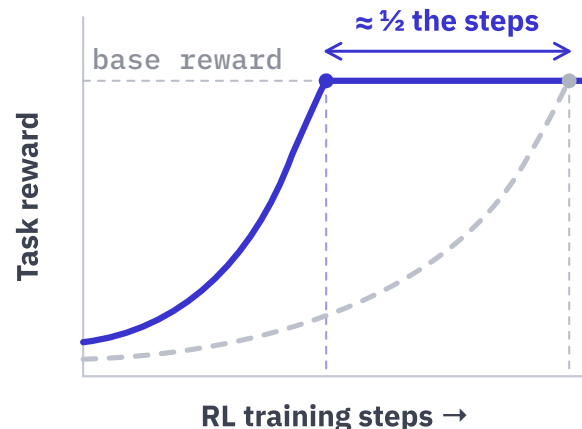


■ Held-out scaffold ■ Trained scaffold

Trained on Mycroft **alone**, the stronger model improves on **all three** scaffolds. Cross-scaffold transfer appears gated by base competence.

Hard-task RL accelerates downstream convergence

init from hard-task RL · vs. cold start



— Init from hard-task RL --- Cold start

Even when the skill doesn't transfer zero-shot, initializing from hard-task RL reaches base reward in **~half the steps** and lifts downstream Pass@8 by **4×**.

Downstream held-out logic · PI-Logic Pass@8

